

Year 10
Mini Test 1.2
Expanding and Factorising



Baldivis
 Secondary College

Name: Answers.

Score: /23
 Total Time: 30 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed.

Expanding and Factorising	
1 Expand $5(2t + 1)$ [1] $= 5 \times 2t + 5 \times 1$ $= 10t + 5 \checkmark$	2 Expand $-3x(x - 4)$ [1] $= -3x \times x - 3x \times -4$ $= -3x^2 + 12x \checkmark$ <i>watch signs</i>
3 Expand and simplify $(x + 7)(x + 1)$ [1] $= x^2 + 1x + 7x + 7 \frac{1}{2}$ $= x^2 + 8x + 7 \frac{1}{2}$	4 Expand and simplify $(3p - 2)(4p + 9)$ [2] $= 3p \times 4p + 3p \times 9 - 2 \times 4p - 2 \times 9$ $= 12p^2 + 27p - 8p - 18 \quad $ $= 12p^2 + 19p - 18 \quad $
5 Expand and simplify $(a - 5)^2$ [2] $(a - 5)(a - 5)$ $= a^2 - 5a - 5a + 25 \quad $ $= a^2 - 10a + 25 \quad $	6 Expand $(2x + 3)^2 - 3(x + 1)$ [3] $(2x + 3)(2x + 3) - 3(x + 1)$ $= 4x^2 + 6x + 6x + 9 - 3x - 3 \quad $ $= 4x^2 + 9x + 6 \quad $

7 Expand and simplify $(3a + 5c)(3a - 5c)$ [2] $= 9a^2 - 15ac + 15ac - 25c^2$ $= 9a^2 - 25c^2$	8 Factorise $x^2 + 16x$ [1] $x^2 = x \times x \quad 16 = 16 \times 1 \times x$ $x(x + 16) \checkmark$
9 Factorise $12p + 6$ [1] $12p = 2 \times 6 \times p \quad 6 = 2 \times 3$ $2(6p + 3) \checkmark$	10 Factorise $3x^2 - 9x$ [2] $3x^2 = 3 \times 1 \times x \times x \quad -9x = 3 \times -3 \times x$ $3x(x - 3)$ 1 for HCF 1 for factored form.
11 Factorise $x(x + 3) + 2(x + 3)$ [1] $= (x + 3)(x + 2) \checkmark$	12 Factorise $a^2 + 4a + 3$ [2] $(a + 3)(a + 1)$ $a^2 + a + 3a + 3$
13 Factorise $a^2 + 4a + 3$ [2] double up $m^2 - 5m + 6$ $(m - 3)(m - 2) \quad 2 \times 3$	14 Factorise $m^2 + 5m - 6$ [2] $a^2 - 36$ $(a - 6)(a + 6) 2$